PLC: Direct Logic DL-06

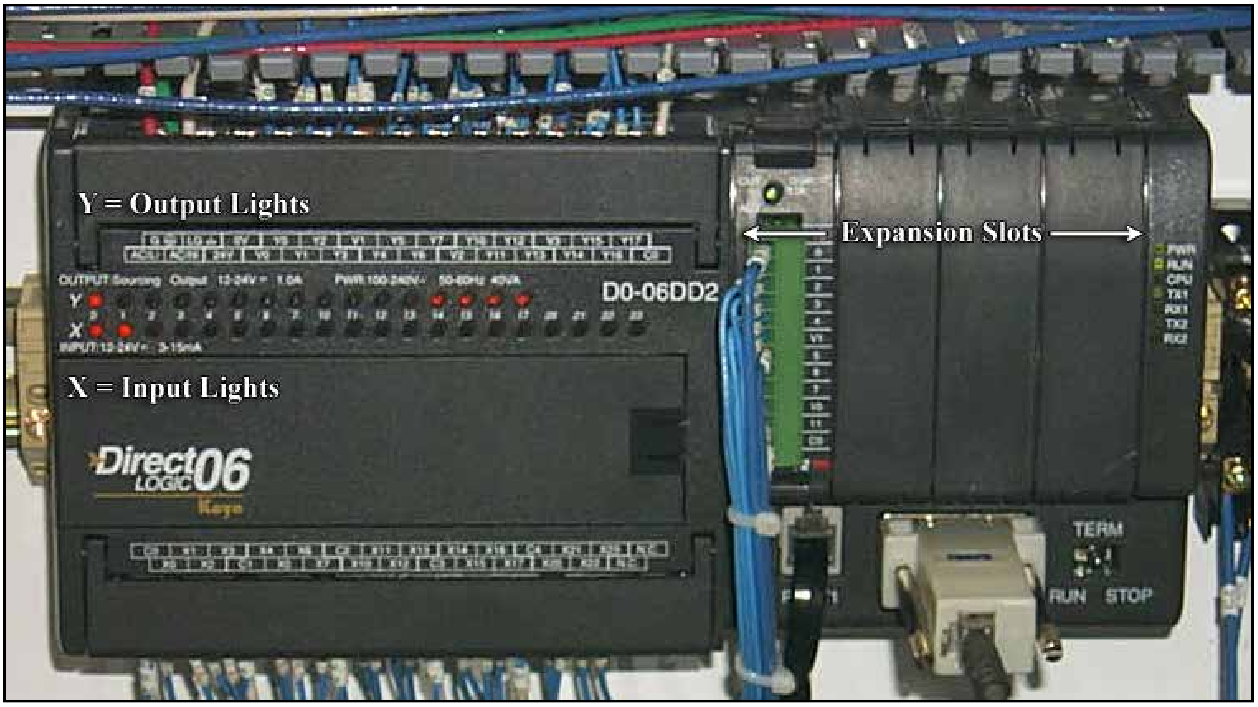

Figure 45-1
Direct Logic DL-06

PLC: Inputs and Outputs

PLC Base / 20-PT DC In, 16-PT DC Out

Connection	Description	Device
X0	Spare	
X1	Spare	
X2	24VDC Safety OK	
X3	Count Sensor 1	PE6041
X4	Count Sensor 2	PE6061
X5	Count Sensor 3	PE6071
X6	Count Sensor 4	PE6081
X7	Count Sensor 5	PE6091
X10	Servo At Target	DR9001
X11	Servo Faulted	DR9001
X12	Servo At Target	DR9141
X13	Servo Faulted	DR9141
X14	Servo At Target	DR9501
X15	Servo Faulted	DR9501
X16	Servo At Target	DR9641
X17	Servo Faulted	DR9641
X20	E-Stops OK	
X21	Guard Closed	
X22	Operator Station Cycle Start	PBLT6581
X23	Operator Station Cycle Stop	PB6591

Figure 46-1
Product Count Photo Eyes

Counter Stacker Rows, Photoeyes, Proximity Sensors, Servo Motors, and Interrupt Forks are numbered with 1 being on the right side as viewed from the infeed.

Connection	Description	Device
Y0	Infeed Conveyor Drive Enable	CR5051
Y1	Discharge Conveyor Drive Enable	CR5061
Y2	Diverter Gate	SOL5071
Y3	Upper Interrupt Fork 1 In	SOL5081
Y4	Upper Interrupt Fork 2 In	SOL5101
Y5	Upper Interrupt Fork 3 In	SOL5111
Y6	Upper Interrupt Fork 4 In	SOL5121
Y7	Upper Interrupt Fork 5 In	SOL5131
Y10	Upper Interrupt Fork 6 In	SOL5511
Y11	Stack Vibration Drive Enable	CR5521
Y12	Audible Alarm, Machine Start	ABU5531
Y13	Cycle Start Light	PBLT6581
Y14	Air Assist Nozzle 1 On	SOL5561
Y15	Air Assist Nozzle 2 On	SOL5571
Y16	Air Assist Nozzle 3 On	SOL5581
Y17	Air Assist Nozzle 4 On	SOL5591

PLC: Inputs and Outputs

Slot Exp. 1 / 16-PT DC In

Connection	Description	Device
X100	Tower 1 Down	PRS7011
X101	Tower 2 Down	PRS7021
X102	Tower 3 Down	PRS7031
X103	Tower 4 Down	PRS7041
X104	Tower 5 Down	PRS7061
X105	Tower 6 Down	PRS7071
X106	Servo At Target	DR10001
X107	Servo Faulted	DR10001
X110	Servo At Target	DR10141
X111	Servo Faulted	DR10141
X112	Spare	
X113	Spare	
X114	Spare	
X115	Spare	
X116	Spare	
X117	Spare	

Slot Exp. 2 / 16-PT DC Out

Connection	Description	Device
Y100	Drive 1 Trigger	CBL7511
Y101	Drive 1 Step	CBL7511
Y102	Reset Linmot Error	
Y103	Spare	
Y104	Drive 2 Trigger	CBL7561
Y105	Drive 2 Step	CBL7561
Y106	Drive 3 Trigger	CBL7581
Y107	Drive 3 Step	CBL7581
Y100	Drive 4 Trigger	CBL7611
Y111	Drive 4 Step	CBL7611
Y112	Drive 5 Trigger	CBL7631
Y113	Drive 5 Step	CBL7631
Y114	Drive 6 Trigger	CBL7651
Y115	Drive 6 Step	CBL7651
Y116	Spare	
Y117	Spare	

PLC: Inputs and Outputs

Slot Exp 3 / 10-PT DC Out

Connection	Description	Device
Y120	Air Assist Nozzle 5 On	SOL8011
Y121	Air Assist Nozzle 6 On	SOL8021
Y122	Lower Interrupt Fork 1 In	SOL8031
Y123	Lower Interrupt Fork 2 In	SOL8041
Y124	Lower Interrupt Fork 3 In	SOL8051
Y125	Lower Interrupt Fork 4 In	SOL8071
Y126	Lower Interrupt Fork 5 In	SOL8081
Y127	Lower Interrupt Fork 6 In	SOL8091
Y130	Spare	
Y131	Spare	

Guard Interlock

GDS6001 Fork Cover Safety Switch

These sensors cause the machine to stop when one of the guard doors has been opened.

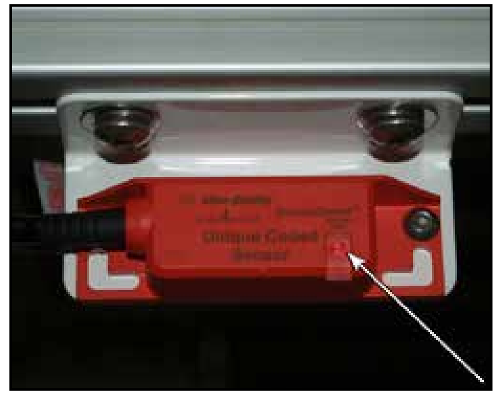

figure 49-1
SensaGuard Sensor

Indicator Diagnostics for Guard Interlock
Unit Indicators (per IEC 60073)

	State	**Status**	**Troubleshooting**
Status / Diag LED	Off	Not Powered	NA
	Red	Not Safe, OSSD not active	NA
	Green	Safe, OSSD active	NA
	Green Flash	Power up test or OSSD inputs not valid	Check 24V DC or OSSD inputs (yellow and red wire)
	Red Flash	1Hz Flash Recoverable Fault 4 Hz Flash Non-recoverable Fault	Recoverable Fault - check OSSD outputs are not shorted to GND, 24V DC or each other. Cycle Power.
	Amber Flash	1 Hz Flash margin indicator safe, OSSD active	Sensor is reaching max. sensing distance; re-align sensor with actuator

Timers and Count Functions

TO OBTAIN A MAXIMUM PERFORMANCE OPERATION OF THE *GENESIS COUNTER STACKER* THESE TIMERS MUST BE SET ACCURATELY BY TRAINED PERSONNEL ONLY.

NOTE: CHECK TO ENSURE AIR PRESSURE IS SET AT 40 PSI BEFORE MAKING TIMER ADJUSTMENTS.

Counter Stacker - Machine Sequence

As product is fed into the counter stacker it is guided into rows or lanes. For each lane:

1. As each product passes beneath the Product Eye a counter is incremented.
2. When the current lane count is equal to the **Upper Fork Hold Count** the upper interrupt forks will retract, dropping the stack to the lower interrupt forks.
3. When the current lane count is equal to the **Lower Fork Hold Count** the lower interrupt forks will retract, dropping the stack to the stacking tower.
4. When the **Step Count** is reached the stacking tower will lower slightly allowing more room for product in the stacking cup. As the stack is accumulated and the stack count reaches the **Second Level Start** percentage the **Second Level Step Count** takes effect. The **Step Count** is ignored. The **Second Level Step Count** will continue to step the stack down until the **Stack Count** is reached.
5. When the current lane count is equal to the desired **Stack Count** for that lane, the **Travel Time** will begin. When the **Travel Time** times out the stacking tower will lower to the red belts, releasing the stack to the Auto Stack Indexer.
6. The **Fork Delay Timers** are started and when they time out the correspoindng interrupt forks will extend to capture the next product.
7. When the stacking tower actuates the Servo Motor Down sensor the **Tower Down Time** starts. When the **Tower Down Time** is reached the stacking tower will rise in preparation to support the next stack.
8. The **Tower Up Time** begins and when complete the interrupt forks may retract (see step 2, above) dropping the next stack onto the stacking towers.

How To Adjust Lane Counts and Timers

There are several timers that can be adjusted to optimize the stacking performance. The operator interface allows the operator access to these timers to allow adjustment.
See pages 27-29 for instructions on how to change the existing values for these timers.

Fork Hold Count

Fork Hold Count controls when the interrupt fork is allowed to retract. To achieve the best quality stack, set the count value to the amount that nearly fills the cup cylinder, but does not exceed the top level before the forks retract.

Step Count

The 'Step Count' is the number of product pieces that will accumulate between tower drops before the Second Level Start. It is very important that as the stack accumulates this plain is maintained just below the interrupt forks.
If the Step Count is set too low, every time a piece passes by the stack height eye, it will stepdown the stacking tower. This condition is not recommended. The stacking tower should step down only after 3-6 pieces have accumulated, depending on thickness. To correct this problem increase the Step Count.
If the Step Count is set too high, an overload condition may occur, causing the stack height to accumulate above the interrupt forks lane of travel. This can cause pinching on the top of the stack when the interrupt forks advance, after the total stack is reached. This will result in miscounts, poor quality stacks and a jam within the stacking cup cylinder. To correct this condition, decrease the Step Count.

Second Level Start

The percentage of the stack to accumulate before the second level starts.

Timers and Count Functions

Step Changeover point
The count, based on Second Level Start at which the system will begin to monitor for the Second Level Step Count.

Second Level Step Count
The number of pieces between tower 'step downs' to maintain the stack height just below the top of the stack cup after the Second Level starts.

Travel Time
The Travel Time is defined as the amount of time it takes for the piece to travel from the count photoeye to the stack. The accuracy of this timer is crucial to the neatness of the final stack.
If the Travel Time is too short the stack will begin to drop down and the interrupt fork will advance into position before the last piece has a chance to land on the stack. This can cause miscounts of the present stack and the subsequent stack, as well as a poor quality stack. To correct this scenario, increase the Travel Time for the number of rows selected.
If the Travel Time is set too long, the interrupt fork will advance into position too late to catch the first piece of the next stack. This can cause miscounts of the present stack and the subsequent stack, as well as a poor quality stack. To correct this scenario, decrease the Travel Time for the number of rows selected.
If the Travel Time is adjusted correctly, the last piece of the present stack will have enough time to settle onto the stack and the interrupt fork will advance into position to collect the first piece of the subsequent stack.

Fork Delay Time
The Fork Delay Time is defined as the amount of time allowed for the completed stack to start down before the interrupt fork advances forward to catch the first tortilla of the subsequent stack. The accuracy of this timer is crucial to the neatness of the stack.
If the Fork Delay Time is too short, the forks will advance forward too early, which may hit the exiting stack or even impale the last piece for this stack. To fix this scenario, increase the Fork Delay Time.
If the Fork Delay Time is too long, the forks can impale the first piece of the subsequent stack when advancing forward or not catch this piece at all. To correct this problem, decrease the Fork Delay Time.
When the Fork Delay Time is adjusted correctly, the last tortilla of the exiting stack will have enough time to settle on the stack and the interrupt fork will advance into position to collect the first piece of the subsequent stack.

Tower Down Time
The down timer is energized when the stacking tower has reached the full down position and the proximity switch has been made. The stacking tower will not return to the full up position until the proximity switch is made and the down time has timed out. The down time is the time that the stacking platform remains in the full down position while the completed stack exits the machine. The accuracy of this timer is also critical to the integrity of the final tortilla stack.
If the Down Time is too short, the stacking tower will return upward prematurely, hitting the back of the exiting stack. This can result in a slightly tipped stack, a scattered pile, or the stack not having enough time to exit the platform and the stack being jammed up into the stacking cup cylinder when the stacking tower returns up for the following stack. To correct this problem, increase the Down Time for the number of rows selected.
If the Down Time is too long, the stacking tower will be delayed in returning to the full up position, which can allow too many pieces for the next stack to accumulate on the interrupt fork. This can result in a poor quality stack as well as a miscount of this stack. To correct this problem, decrease the Down Time for the number of rows selected.
If the Down Time is set properly, the exiting stack will have enough time to advance away from the stacking tower, while allowing the tower to return to its full up position before an overload condition occurs on the interrupt fork.

Tower Up Time
The Stack Up timer allows for the stacking towers to return to the full up position before the interrupt forks are energized to retract, releasing the accumulated tortillas for the next stack onto the tower. This timer effects the neatness of the stack and the accuracy of the stack count.
If the Stack Up Time is too short, the towers will not reach the full up position. The interrupt forks will then retract and

Timers and Count Functions

release the next stack. With this increased distance between the interrupt forks and the tower the next stack will be dropping too far and cause a poor quality stack. To correct this problem, increase the Stack Up Time.
If the Stack Up Time is too long, the interrupt forks will be delayed from retracting and too many pieces will accumulate on the forks. This can cause a miscount as well as a poor quality stack. To correct this scenario, decrease the Tower Up Time.
When this timer is set correctly, the stacking tower will travel to the full up position and the interrupt forks will retract when the fork hold count is reached. This is the optimum setting and will provide the best performance and quality stack.

Sensor Max Time
PE Blocked Time is the maximum amount of time the Count PE is expected to be blocked. If this timer times out an Auto Clear Out Cycle will take place for the corresponding lane. The Interrupt Forks will retract and the stacking tower will drop. They will remain in this position until the Count PE is no longer blocked.

Debounce Option
Debounce Time
Sensor Debounce Time is the amount of time after which the count sensor is triggered during which any further input from the sensor is ignored. This helps prevent an individual piece of product from being counted more than once due to rapid on/off/on transitions of the sensor as the product travels under it.

Minimum On Time
Sensor Minimum Time is the minimum amount of time required for count to increase when the sensor is activated.